

Let there be...hacks!!!

SENTRY

Ioannis Zografopoulos, Alyah Alfageh, Li Zhou, Shijie Pan

Hack My Robot Competitions – CSAW 2022

Secure Next Generation Resilient Systems Lab

sentry.kaust.edu.sa



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King Abdullah University of
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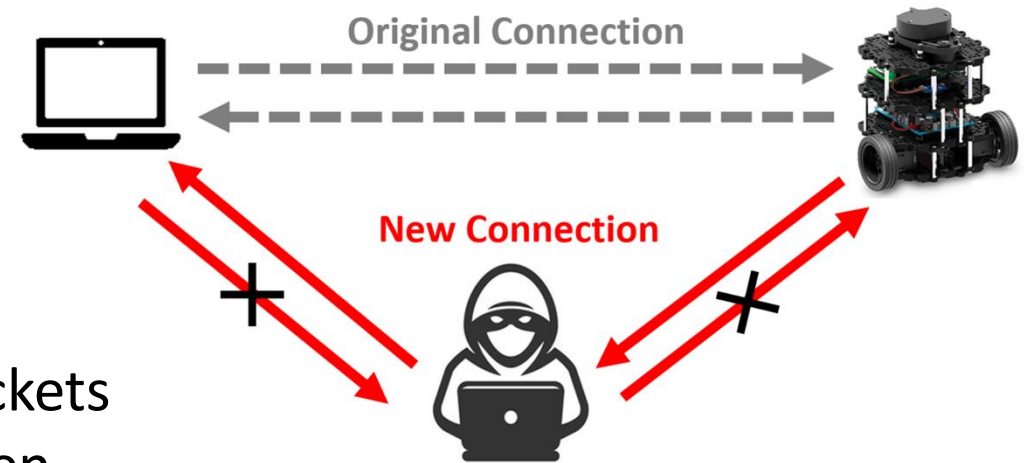
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SECURE NEXT GENERATION RESILIENT SYSTEMS LAB



Compromising Availability 1/2

- Attack: ARP* poisoning
- Methodology
 - Send forged ARP packets
 - Deceive communication endpoints
 - Become Man-in-the-Middle node
 - Intercept, forward, and modify certain packets
 - **Flood the communication** channel between robot and user/server
- Adversarial access: remote – access to network
- Impact: Denial-of-service
- Tools: Arpspoof

*ARP: address resolution protocol



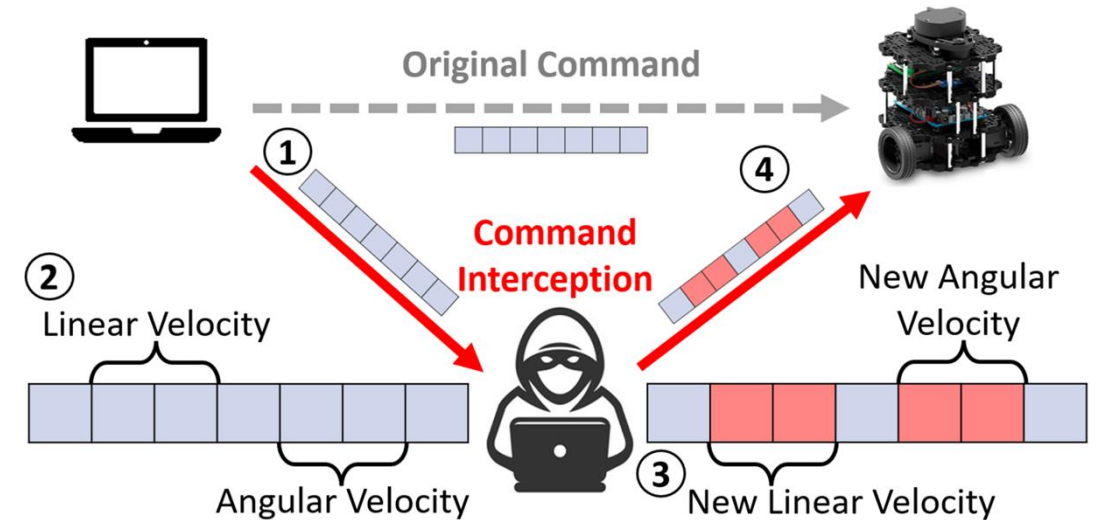


Compromising Availability 2/2

- Attack: Deauthentication
- Methodology
 - Flood communication channel
 - Force the de-authentication of the host (robot)
 - Capture control packets attempting the re-authentication
 - Inspect packets and identify wifi passwords
- Adversarial access: remote – access to network
- Impact: Denial-of-service
- Tools: Kali Linux, airmmon-ng

Taking Control

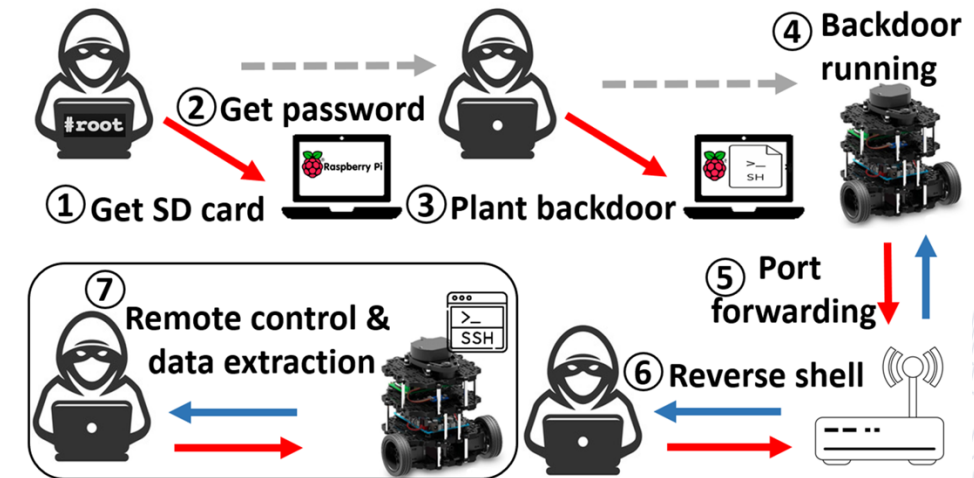
- Attack: Packet modification
- Methodology
 - Successful ARP spoofing attack
 - Capture control packets
 - Modify packet contents
 - e.g., robot's linear/angular velocity
 - Forward maliciously crafted packets to target
- Adversarial access: remote – access to network
- Impact: Loss of controllability, denial-of-service
- Tools: Arpspoof, Scapy, Wireshark, iptables, nfqueue





Becoming Stealthy 1/2

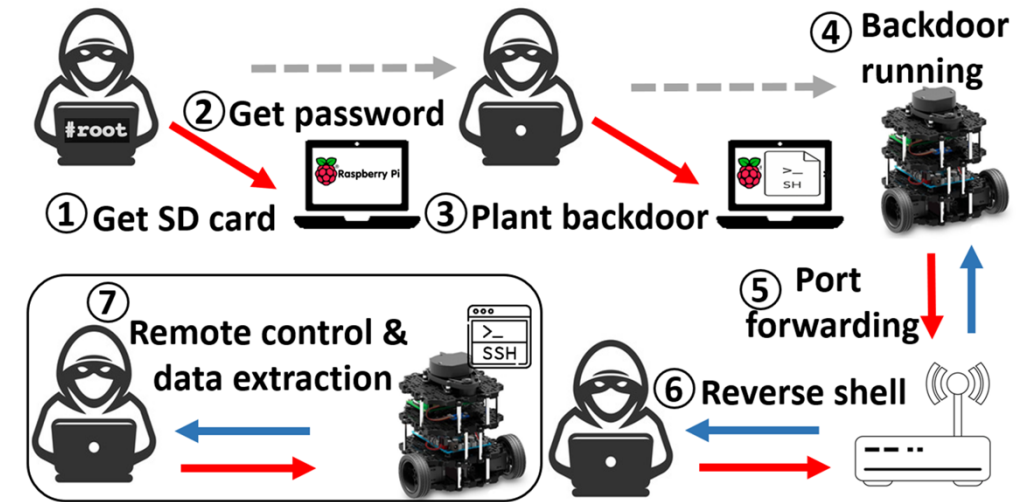
- Attack: Planting a backdoor
- Methodology
 1. Access the SD card and change the boot process
 2. Attain root password for the robot
 - E.g., admin privileges
 3. Plant a backdoor on host using Metasploit
 4. Automatically start reverse shell on reboot
 5. Attain **persistence** and forward communication to malicious user
 - Even if the robot's IP is modified
- Adversarial access: **one-time** physical access
- Impact: Remote control of agent
- Tools: Kali Metasploit, meterpreter, frp





Becoming Stealthy 2/2

- Attack: Data Corruption
- Methodology
 - Attain reverse shell on robot
 - Steps 1 - 6
 - Execute script that
 - Detects image path of the robot
 - Delete, replace, edit, or upload images to host
 - E.g., new images can be detected and removed, malware can be appended to images, etc.
- Adversarial access: **one-time** physical access
- Impact: Remove, modify/corrupt, upload malware
- Tools: Kali Metasploit, meterpreter, frp





Thank you!

Contact us:

